



University of Sri Jayewardenepura- Faculty of Applied Sciences
Department of Mathematics
BSc Degree First Year First Semester Course Unit Examination-November 2022

MAT 112 2.0 Differential Equations

Time : Two (02) hours

Answer all (3) questions.

This paper carries total of 100 marks.

Usual notations apply throughout the paper.

Question 1

(a) Find the order and the degree of the following differential equations.

(i) $\frac{d^2y}{dx^2} = \left[1 + \left(\frac{dy}{dx}\right)^2\right]^3$ (ii) $y = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}}$ (iii) $\frac{d^4y}{dx^4} + \cos \frac{d^2y}{dx^2} = 5x$

(b) Find the critical points of the following differential equation. Sketch the direction fields and discuss the stability.

$$\frac{dx}{dt} = \alpha(x^2 - 5x + 6), \quad \alpha \text{ is a non zero real parameter.}$$

(c) Solve the following differential equations.

(i) $\frac{dy}{dx} = e^{x+y} + x^2 e^y$ (ii) $(x + \sin y)dx + (x \cos y - 2y)dy = 0$
 (iii) $\frac{dy}{dx} - \frac{3y}{1+x} = (x+1)^4$

(d) Solve the following initial value problem.

$$-ydx + (x + \sqrt{xy})dy = 0 ; \quad y(1) = 1$$

(e) Using Wronkian, show that every solution of the following equation is a linear combination of independent solutions of $e^x \cos x$ and $e^x \sin x$.

$$y'' - 2y' + 2y = 0$$

Marks:45

Question 2

(a) Find the set of orthogonal trajectories of the family of curves $\frac{x^2}{16} + \frac{y^2}{\alpha+4} = 1$, where α is a given parameter.

(b) Solve the following differential equation.

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 3y = \sin x + xe^x$$

(c) Solve the following system of differential equations.

$$\frac{dx}{dt} + y = e^t$$

$$\frac{dy}{dt} - x = e^{-t}$$

Marks:35

Question 3

(a) For the following differential equation, classify the singular points.

$$x^2(x^2 - 1)^2y'' - x(x - 1)y' + 2y = 0$$

(b) Obtain the partial differential equation by eliminating a and b from

$$Z = (x^2 + a)(y^2 + b). \text{ Here } Z = Z(x, y).$$

(c) Given that $D = \frac{\partial}{\partial x}$ and $D' = \frac{\partial}{\partial y}$, solve the following partial differential equation.

$$(D^3 - D'^2)z = \sin(2x + 3y)$$

Marks:20

Handwritten note:
 $z = 0 (x + y)$